

Thomas Sawyer

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Professional Summary

Software engineer and technical generalist with experience in internal tools, production automation, developer tooling, and manufacturing systems. Longtime polyglot programmer comfortable becoming productive in unfamiliar languages and systems directly from technical documentation. Works across industrial and software environments, owning projects from design and implementation through integration, debugging, and delivery.

Professional Experience

National Aperture, Inc. | Laser Manufacturing, Software Development & IT | 2018-2025

- Manufactured microscopic precision features using laser micromachining equipment and computer-controlled motion systems.
- Built internal software for production scheduling, reporting, and manufacturing visibility.
- Worked across software development, operations, troubleshooting, and practical IT support in a small manufacturing environment.
- **NAI Schedule**: Internal operations software in **Elixir**, **Phoenix**, **MySQL**, and **Ecto** for Fishbowl-integrated production scheduling, search, and order visibility.
- **Keyence CoA**: Report automation tool in **Elixir** and **Phoenix LiveView** that turned Keyence CSV exports into print-ready Certificates of Analysis in seconds.
- **BENCHCON 2**: Refactor of a legacy laser and motion-stage control system in a **FORTH-variant** production automation environment to improve modularity, units handling, maintainability, and onboarding.

Tab Computing | Independent Product Developer | 2025-Present

- Building an independent portfolio of early-stage SaaS products spanning entertainment, identity and authentication infrastructure, job-search tooling, and AI-assisted marketplaces.

Tab Computing | Freelance Web and Open-Source Developer | 2009-2017

- Built websites and web features for clients.
- Developed open-source applications and libraries, primarily in Ruby.

Software Portfolios

- **Personal GitHub** | <https://github.com/trans>
Personal engineering work and experimentation across AI, developer tooling, systems, and Crystal/Elixir software.
- **Rubyworks** | <https://github.com/rubyworks>
Long-running portfolio of open-source Ruby libraries and developer tools, including Ruby Facets and QED.
- **C0 DATA** | <https://github.com/c0data>
Data-format research and implementations using ASCII control codes for compact, high-throughput structured data.
- **Tab Computing** | <https://github.com/tabcomputing>
Independent software development practice for client and product work, with open-source components supporting a broader body of primarily private projects.

Publications and Preprints

- **Prefix-Trie Factorization of Autoregressive Gradient Descent** | **Preprint** on gradient-factorized autoregressive language-model training over a prefix trie, with a memory-scalable **CUDA** implementation and empirical results showing about **10x fewer optimizer steps** at matched context on Shakespeare. Preprint: <https://github.com/trans/microgpt/blob/main/docs/agpt-paper.md>
- **Trie Path Probability Convergence** | **Preprint** on prefix-trie path probabilities as stable ordinal fingerprints of corpus distributions, with empirical results showing **Spearman rho ≥ 0.80** across depths up to 16 on corpora from **1.1M to 5M characters**. Preprint: <https://github.com/trans/microgpt/blob/main/rnd/results-extended.md>

Skills

- Languages: Crystal, Elixir, Ruby, JavaScript, SQL, C, C++
- Additional Language Experience: Forth, Lisp, Prolog
- Frameworks: Phoenix, LiveView, Ecto, Kemal
- Data: MySQL, PostgreSQL, SQLite; schema design, migrations, and backups
- Infrastructure & DevOps: Docker, Docker Compose, Linux administration, shell scripting, SSH, cron, systemd, DNS, Nginx/Apache, web application deployment, CI with GitHub Actions
- Engineering: Internal tools, automation, parsers and data formats, systems integration, automated testing, troubleshooting, package and release management
- Environment: Linux, Windows, Git, GitHub

Education

Continuing Technical Study | 2025-Present

- Self-directed research and implementation in language-model training, information retrieval, data formats, and multi-agent software systems.

St. John's College, Santa Fe Campus | 1998-2000

- Coursework in Philosophy and History of Mathematics and Science

Anne Arundel Community College | 1992-1993

- Coursework in Computer Science and General Curriculum